Measuring Orbital and Spin periods of Cataclysmic Variables

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The orbital and spin periods of MU-Camelopardalis intermediate polar cataclysmic variable were obtained from the 519 flux measurements across multiple days by the Durham Astrolab telescopes. Two characteristic periods were revealed in the power spectrum via the application of the Lomb-Scargle periodogram on the lightcurve dataset. The orbital period was determined as $P_o = 4.722 \pm 0.053$ hours which is consistent with the literature value of 4.707 hours deviating by 0.29σ. The spin period of the white dwarf was determined as $P_s = 1186.8 \pm 0.7$ seconds which is also consistent with the literature value of $P_s = 1187$ seconds in the range 0.01σ. The results confirm that MU-Cam is an intermediate polar and illustrate that the Lomb-Scargle algorithm is an effective tool for identifying periodic signals in unevenly sampled photometric data.

I. INTRODUCTION

Most of the stars in the universe form in binaries and they are bound by the gravitational forces[1]. During their evolution, the star with the larger mass evolves off the main sequence into a white-dwarf and the companion star remains in the main sequence. This is because the more massive star in the binary system expands and transfers mass to the companion star via Roche-Lobe overflow which leads the massive star to contract into a dense and compact remnant. Cataclysmic variables (CV) are such systems where the compact remnant is a white-dwarf and the companion star is a main-sequence donor. The mass transfer in CVs are carried over gigayear timescales [2] and the transferred mass that has angular momentum forms an accretion disk around the white-dwarf (WD) .Through this process, the gravitational potential energy is converted into radiation emission by the accretion disk, hence CVs have such high luminosities and allow us to investigate their optical properties over time.

CVs that have white-dwarfs with strong magnetic fields ($B \gtrsim 1\,\mathrm{MG}$) are classified as magnetic CVs and they are further subdivided by the magnetic field strengths of the white dwarfs [3].

Polars are magnetic CVs with extremely strong magnetic fields ($B \gtrsim 10\,\mathrm{MG}$) and the WD magnetically locks the orbital period with its spin period due to the strength of WD's magnetic field. Consequently, material flows directly onto the magnetic poles along the magnetic field lines without the formation of the accretion disk.

Intermediate polar (IP) is another type of magnetic CV with a weaker magnetic field compared to the polars ($B \lesssim 10\,\mathrm{MG}$), so an accretion disc still forms but is truncated at the magnetospheric radius, inside which the material is transferred along the field lines to the magnetic poles of the WD similarly to polars [3, 4].

However, the WD spin is not locked to the orbit in IPs due to its insufficient magnetic field strength, thus the IPs produce two distinct detectable signals rather than one. IPs have a distinct orbital period P_{orb}, from geometric modulations of the accretion disc and the companion star as two stars orbit each other. IPs also have a distinct spin period P_{spin} from the WD, produced by the rotating accretion column which in turn illuminates the disc and donor at this spin frequency.

These two distinct frequencies are the observational signatures of intermediate polars that can be simultaneously identified.

MU-Camelopardalis ($\alpha = 06^{\mathrm{h}}25^{\mathrm{m}}16.2^{\mathrm{s}}$, $\delta = +73°34'40''$, J2000) is a confirmed IP, identified by its strong X-ray emission in optical spectroscopic studies of X-ray sources [5].

Further photometry research on MU-Cam calculated an orbital period of $P_{orb} = 4.707$hr and a WD spin period of $P_{spin} = 1187$s (≈ 19.78 min), placing MU-Cam among the known intermediate polars in the catalogue of Ritter & Kolb [6].

Identifying periodic signals in unevenly sampled photometric data requires a Fourier method robust to unevenly sampled data.

The Lomb–Scargle (LS) periodogram [7, 8] provides this by testing, at each trial frequency f, how well a oscillating function of the form

$$x(t) = A\cos(2\pi f t) + B\sin(2\pi f t), \tag{1}$$

fits the data where A and B are free amplitude coefficients [9].

The normalised power $P(f)$ is computed for the y-axis where the magnitude of $P(f)$ indicates the frequency signals in the data. The frequency resolution $\Delta f \approx 1/T$ sets the minimum separation between these frequency peaks, and the Nyquist frequency

$$f_{\mathrm{Nyq}} = \frac{1}{2\,t_{\mathrm{exp}}}, \tag{2}$$

where t_{exp} is the exposure time, gives the upper frequency bound in for the frequency grid where beyond this frequency, signals cannot be reliably calculated [9].

The objective of this investigation is to measure the orbital and spin periods of MU-Cam by computing a Lomb-Scargle periodogram of the light curve data obtained by the Durham University's Astrolab telescopes. After extracting the f_{orb} and f_{spin} values and computing their corresponding uncertainties, the obtained frequencies will be converted to orbital and spin periods to be then compared with the accepted literature values.

II. METHODS

Four different photometry datasets of MU-Cam ($\alpha = 06^{\mathrm{h}}25^{\mathrm{m}}16.2^{\mathrm{s}}$, $\delta = +73°34'40''$, J2000) were obtained using three Durham AstroLab telescopes across 4 different dates. The properties of each dataset are summarised in Table I.

The differential magnitude of MU-Cam relative to the ensemble mean of the comparison stars was extracted for each frame together with its photometric uncertainty σ_m and in

TABLE I: All available photometry datasets for MU-Cam are summarised below. Datasets labelled "used" were combined for the primary analysis

Telescope	Date (UT)	Filter	$t_{\rm exp}$ (s)	Used
East-16	2025 Feb 24	C	60	Yes
Draco-2	2025 Feb 25	V	60	Yes
East-14	2022 Jan 07	C	30-60	No
East-14	2022 Jan 11	C	15	No

Julian Date (UTC).

$$m_{\rm diff} = m_{\rm target} - \langle m_{\rm ref} \rangle, \tag{3}$$

Multi-aperture module in the AstroImageJ software was used to apply differential photometry on the raw frames to obtain a lightcurve [10]. The light curves of the chosen comparison stars were inspected independently and the flux variability being uncorrelated confirmed that the flux changes for MU-Cam were due to its astrophysics properties rather than instrumental or atmospheric effects.

To maximise the number of usable measurements while preserving a uniform Nyquist frequency, East-16 and Draco-2 observations were chosen to be combined for the primary analysis of MU-Cam's orbital and spin periods. This is because these datasets were obtained in consecutive days with a common exposure time of 60s, giving us the same Nyquist limit of $f_{\rm Nyq} = 720$ cycles day^{-1}.

Combining datasets with differing exposure times would give us differing Nyquist frequency across the combined dataset, introducing additional complexity in the analysis of the power spectrum (see Discussion).

The East-16 observation produced 57 measurements over $\sim$1.9 hours , while the Draco-2 observation yielded 462 measurements over $\sim$8.4 hours.

Combining these two time-series data produced a combined dataset of $N = 519$ measurements with a total baseline of $T = 1.36$ days with a gap of 22 hours in between the measurements.

Prior to period analysis, all differential magnitudes were converted to proportional fluxes via

$$F = 10^{-m_{\rm diff}/2.5}, \tag{4}$$

which linearises the photometric scale as required for least-squares sinusoidal fitting during the Lomb-Scargle analysis.

Firstly, The Lomb-Scargle (LS) periodogram was computed in Python using the astropy.timeseries.LombScargle seperately for each of the four individual datasets to verify that the same peaks were observed. The Lomb-Scargle algorithm involves floating-mean generalisation to account for any DC offset in the flux series.

For each periodogram, the frequency grid spanned from $f_{\rm min} = 2/T_i$ (where T_i is the individual baseline) to $f_{\rm max} = 86400/(2\,t_{\rm exp})$ cycles day^{-1}, sampled at 1000 uniformly spaced points.

The same way, the primary Lomb-Scargle periodogram was computed for the combined East-16 and Draco-2 dataset to identify the orbital and spin frequencies of MU-Cam.

This analysis involved a two step process, a broad grid spanning $f_{\rm min} = 2/T \approx 1.48$ cycles day^{-1} to $f_{\rm max} = 720$ cycles day^{-1} at 10,000 points was used to investigate the full accessible frequency range and identify the locations of the dominant peaks.

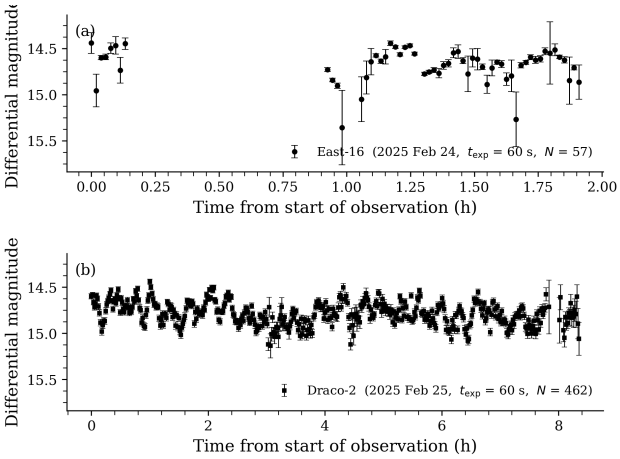

FIG. 1: Differential light curve of MU-Cam from the two observations used in the primary analysis.(a) East-16 telescope, 2025 February 24 ($N = 57$, $t_{\rm exp} = 60$ s), the 48-minute gap near $t = 0.1$ h indicates an interruption in the observation. (b) Draco-2 telescope, 2025 February 25 ($N = 462$, $t_{\rm exp} = 60$ s). The time axis in each panel is measured from the first frame of that observation. Error bars show 1σ uncertainties (derived from Poisson distribution) on the source and background noise. Brighter flux corresponds to a smaller differential magnitude due to the inverted magnitude axis.

Then, the frequency grid was adjusted to a narrower window centred on each candidate signal — $[3,\ 8]$ cycles day^{-1} for the orbital frequency peak and $[60,\ 80]$ cycles day^{-1} for the spin frequency peak.

The frequencies with the highest power in each window was recorded as the measured value of $f_{\rm orb}$ and $f_{\rm spin}$ respectively, and the periods were computed using the equation $P = 1/f$.

Lastly, the errors on the peak frequencies were deduced by applying Monte Carlo bootstrapping, full details of the error estimation are given in the Error Appendix.

III. RESULTS

The differential light curves of MU-Cam are shown below in Fig.1 illustrating the periodic change in flux over time for the primary datasets Draco-2 and East-16.

Differential magnitudes were converted to proportional fluxes using the formula $f = 10^{-m/2.5}$ then a Lomb-Scargle periodogram was computed for the combined dataset, shown in Fig. 2. The power spectrum detected two statistically-significant peaks for orbital and spin frequencies consistent with the known values for MU-Cam.

The measured frequencies and periods derived from them are compared to the literature values in Table II. Both signals are within 0.3σ of the literature values.

IV. DISCUSSION

Physical interpretation and consistency with literature

The orbital and spin periods of MU-Cam were successfully calculated and the peak frequencies in the peri-

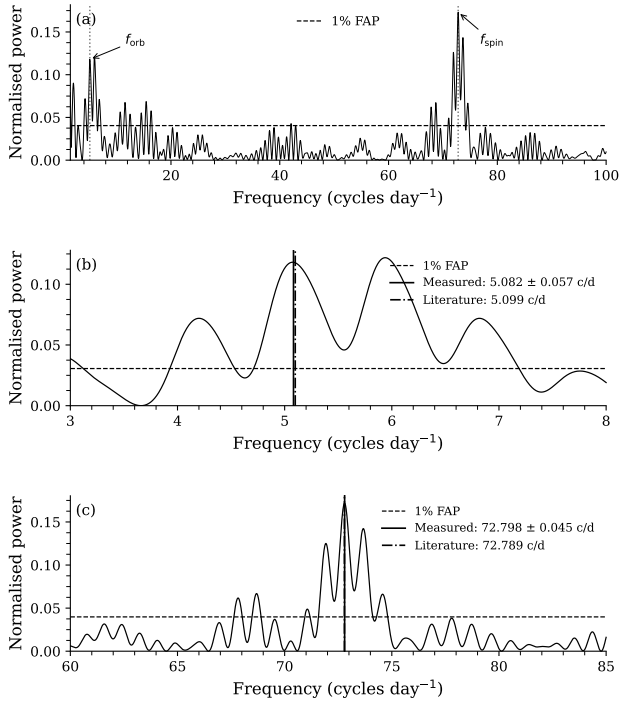

FIG. 2: Lomb–Scargle periodogram of the combined dataset (East-16 and Draco-2) with $N = 519$, baseline $T = 1.36$ days). (a) Full power spectrum over 1.5–100 cycles day^{-1}, with orbital and spin signals labeled. (b) Closer view of the orbital signal where the solid vertical line marks the peak frequency $f_{\rm orb} = 5.082 \pm 0.057$ cycles day^{-1} and the dotted line marks the literature value [6]. (c) Closer view of the spin signal $f_{\rm spin} = 72.798 \pm 0.045$ cycles day^{-1} compared with the literature value in the same way. The horizontal dashed line in each panel is the 1% false-alarm probability (FAP) threshold and both peaks exceed it showing the statistical significance of the signals.

TABLE II: Measured orbital and spin frequencies and periods of MU-Cam compared to the literature values. Measured frequencies are the means of the Monte Carlo bootstrap distributions (see Error Appendix) periods are obtained from the equation $P = 1/f$ with uncertainties propagated as $\sigma_P = \sigma_f/f^2$. The literature values are taken from [6]. The discrepancy z is expressed in units of the measured 1σ uncertainty.

Quantity	Measured	Literature	Discrepancy
$f_{\rm orb}$ (cycles day^{-1})	5.082 ± 0.057	5.099	$0.30\,\sigma$
$P_{\rm orb}$ (hr)	4.722 ± 0.053	4.707	$0.29\,\sigma$
$f_{\rm spin}$ (cycles day^{-1})	72.798 ± 0.045	72.789	$0.22\,\sigma$
$P_{\rm spin}$ (s)	1186.8 ± 0.7	1187	$0.01\,\sigma$

odogram exceed the 1% false alarm probability threshold (Fig. 2) which confirms that the detected peaks are statistically significant and are not due to random noise in the light curve data.

The measured orbital frequency, $f_{orb} = 5.082 \pm 0.057$ cycles day^{-1}, is consistent with the literature value of 5.099 cycles day^{-1} within 0.30σ (Table II). The measured spin frequency, $f_{spin} = 72.798 \pm 0.045$ cycles day^{-1}, also agrees with the literature value of 72.788 cycles day^{-1} to within 0.22σ. It can be concluded that the final results are statistically consistent with the accepted literature values as both results deviate under 1σ.

Additionally, MU-Cam's classification as an intermediate polar is confirmed by our investigation as detecting two independent periodic signals one at $P_{orb} \approx 4.7$ hr and one at $P_{spin} \approx 1187$ s is the observational signatures of an intermediate polar binary system (as outlined in the Introduction).

The orbital modulation observed arises from the orbiting motion of the two stars where the visible area of the accretion disc and donor changes with the viewing angle, thus causing the measured flux to periodically rise and fall. This sinusoidal variation in flux can be seen in Fig. 1. The spin modulation observed arises from the the magnetically channelled accretion column, that rotates with the white dwarf and sweeps across the disc and donor surface with the spin period thus we can observe the varying flux at the spin frequency [3, 4].

The measured spin frequency is larger than the orbital frequency by approximately a factor of 14. Thus the relation $f_{spin} \gg f_{orb}$ confirms that for this binary star system, the white dwarf spins asynchronously with the orbit which is a defining characteristic for an IP system where the magnetic field strength is insufficient to magnetically lock the orbital period with its spin period[3].

Comparison with the literature method

The accepted orbital and spin periods of MU-Cam were measured by larger-aperture telescopes over longer and continuous baselines compared to the measurements we obtained from the Durham University AstroLab telescopes [5]. Despite the inferior infrastructure and quality of the datasets, this investigation managed to extract the same orbital and spin signals of MU-Cam within 0.3σ deviation from the accepted results. This shows that the application of the Lomb-Scargle algorithm is a robust and successful method for extracting the desired astronomical signals when working with a non-uniform photometric dataset with a short baseline.

Another method that is used in converting time-series flux data into a frequency domain is the standard discrete Fourier Transform (FT) which would successfully extract the orbital and spin frequencies if the dataset were to be uniformly sampled. Due to the 22-hour gap between East-16 and Draco-2 observations, LS periodogram was preferred over the standard FT which was successful in extracting the signals without any interpolation in the dataset and thus no artificial frequencies arising from data imputation were detected [7, 8].

Sources of uncertainty in the results

One source of uncertainty in our flux measurements is due to atmospheric scintillation. There are refractive fluctuations in the atmosphere caused by changing air temperature and density which causes the telescopes to capture fluctuating noise in the photometric observations. This random noise affects the differential magnitude measurements in the plotted light curves as seen in Fig. 1 and consequently it produces less precise frequencies in the LS periodogram by raising the noise floor. These atmospheric fluctuations are classified as high frequency noise and the effect of this

scales approximately as $t_{\rm exp}^{-1/2}$ [11]. To improve our results, doubling the exposure time to 120 s would be an effective way to reduce this scintillation noise by a factor of $\sqrt{2}$ to improve the precision of the frequencies in the periodogram.

The resampled datasets during the Monte-Carlo uncertainty estimation resampled the $N = 519$ measurements independently with replacement. However, consecutive photometry frames tend to share correlated observational noise due transparency of the atmosphere and seeing which is not removed by differential photometry if the resampling is done independently for each data point. This means that all observations cannot be treated as statistically independent so the effective number of independent measurements is likely less than $N = 519$, thus the bootstrapping method likely underestimates σ_f to some degree that is seen in Table II.

Updating the bootstrap resampling method by resampling adjacent blocks of data points instead of resampling each data point individually in the combined dataset would help preserve the correlations in the frames. This approach will account for the correlated noise more accurately, thus giving a more accurate uncertainty for our results [12].

Effect of the data selection on the extracted frequencies

For the primary analysis, only the East-16 and Draco-2 datasets were combined, giving us a total baseline of $T = 1.36$ days and a frequency resolution of $\Delta f \approx 1/T \approx 0.74\,\text{cycles day}^{-1}$.

The two 2022 East-14 datasets were excluded because their exposure times (15s and 30s) differed from those of the 2025 datasets, and there was a considerable gap (3 years) between the datasets. This would introduce a non-constant Nyquist frequency in the primary dataset due to the differing exposure times. Furthermore, the ≈3-year gap between the datasets would introduce severe aliasing. Due to the gap in the time-series data, LS cannot distinguish a real signal at frequency f_0 from a false signal (alias) at $f_0 \pm 1/\Delta t$, because the sine function fitting is done equally well across the interruption. A gap of $\Delta t \approx 1000$ days places alias signals at $f_0 \pm 1/\Delta t \approx f_0 \pm 0.001\,\text{cycles day}^{-1}$ which would place the false signals very close to the real frequency peaks making it more difficult to identify the real peak. This is the primary reason the 2022 datasets were excluded from the primary analysis. However, for the 22-hour gap in the combined dataset ($\Delta t \approx 0.93$ days), this spacing is $1/0.93 \approx 1.08\,\text{cycles day}^{-1}$, placing aliases well away from the real peaks making the real signal identifiable.

Implications of the investigation

The calculation of P_{orb} and P_{spin} for MU-Cam to 0.3σ of their literature values using a 1.36 day dataset validates both the observational methodology and the utility of the Lomb-Scargle algorithm. The independent detection of these two periods with photometry without the need for X-Ray observations confirms that MU-Cam is an IP. This demonstrates that we can meaningfully contribute to the characterisation of magnetic CVs even with smaller aperture telescopes observing for short baselines.

The results are consistent with the Ritter Kolb catalog, where P_{spin}/P_{orb} ratios for IPs are in the range 0.01-0.1. This is because the value for MU-Cam $P_{spin}/P_{orb} = 1187\,\text{s} / (4.72 \times 3600\,\text{s}) \approx 0.070$ falls within this range [6].

V. CONCLUSIONS

The orbital and spin periods of the IP MU-Cam were measured from the photometry time-series data obtained by the Durham AstroLab telescopes. Lomb-Scargle periodogram was computed from the 519 flux measurements from two consecutive nights, which yielded $P_{orb} = 4.72 \pm 0.05$ hr and $P_{spin} = 1187.0 \pm 0.7$ s, consistent with the accepted literature values [6] to within 0.30σ and 0.22σ respectively. The simultaneous detection of orbital and spin periods confirm the IP classification of MU-Cam and shows that Lomb-Scargle algorithm can extract these signals to a high precision from an unevenly sampled dataset with a short observational baseline.

The primary limitations were the short duration of the baseline of $T = 1.36$ days, which set a frequency resolution of $\approx 0.74\,\text{cycles day}^{-1}$and the necessary exclusion of the earlier 2022 data due to the severe aliasing caused by the 3-year gap.

In the future, observations with a longer baseline, constant exposure times, and a larger aperture telescope would improve the precision of the measured frequencies and reduce the bootstrap uncertainties on both periods.

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ERROR APPENDIX

AstroImageJ software automatically calculates a 1σ photometric uncertainty σ_m on the differential magnitude of each frame. This error on the magnitude is derived from Poisson distribution applied to the source and background flux counts. These uncertainty values are recorded as one of the columns in the datasets produced by differential photometry.

Differential magnitudes are converted to proportional fluxes using the equation below.

$$F = 10^{-m/2.5}. \tag{5}$$

Differentiating with respect to m gives

$$\frac{\mathrm{d}F}{\mathrm{d}m} = -\frac{\ln 10}{2.5}\, 10^{-m/2.5} = -\frac{\ln 10}{2.5}\, F, \tag{6}$$

so the propagated flux uncertainty for each frame is

$$\sigma_F = \frac{\ln 10}{2.5}\, F\, \sigma_m. \tag{7}$$

These calculated flux uncertainties are used as the vertical error bars for each data point shown in Fig. 1.

For the Lomb-Scargle periodogram, the uncertainties on the peak frequencies f_{orb} and f_{spin} were estimated by Monte Carlo methods and bootstrap resampling techniques.

The combined dataset of $N = 519$ measurements was resampled with replacement $N_{\mathrm{iter}} = 3000$ times. For each sample with the replacement dataset $\{f^{(i)}\}$, the Lomb –Scargle algorithm was applied to produce a periodogram, and the frequency corresponding to the maximum power $f^{(i)}_{\mathrm{peak}}$ for each sampled dataset was saved by adjusting the frequency grid to zoom into a closer range of the signal of interest. The 1σ uncertainty on the peak frequency is taken as the standard deviation of the resulting distribution of the saved peak frequencies.

$$\sigma_{f_0} = \left[\frac{1}{N_{\mathrm{iter}} - 1} \sum_{i=1}^{N_{\mathrm{iter}}} \left(f^{(i)}_{\mathrm{peak}} - \bar{f}\right)^2\right]^{1/2}, \tag{8}$$

where $\bar{f}$ is the mean of the bootstrap distribution. The measured frequency is taken as $\bar{f}$. Figs. 3 show the bootstrap distributions for f_{spin} and f_{orb} respectively, together with the fitted Gaussian and the $\pm 1\sigma$ interval.

The apparent Gaussian Distribution of the saved peak frequencies confirm that the standard deviation σ is an appropriate measure of the uncertainty.

The orbital and spin periods are calculated from their corresponding frequencies from the relation $P = 1/f_0$.

Differentiating gives $\mathrm{d}P/\mathrm{d}f_0 = -1/f_0^2$, so by first-order propagation,

$$\sigma_P = \frac{\sigma_{f_0}}{f_0^2}. \tag{9}$$

For the orbital period expressed in hours (f_0 in cycles day^{-1}),

$$\sigma_{P_{\mathrm{orb}}} = \frac{24\, \sigma_{f_{\mathrm{orb}}}}{f_{\mathrm{orb}}^2}. \tag{10}$$

For the spin period in seconds (f_0 in cycles day^{-1}),

$$\sigma_{P_{\mathrm{spin}}} = \frac{86400\, \sigma_{f_{\mathrm{spin}}}}{f_{\mathrm{spin}}^2}. \tag{11}$$

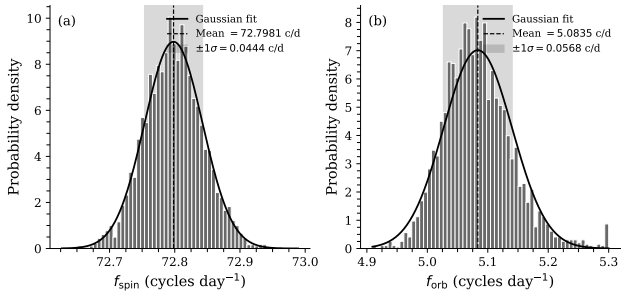

FIG. 3: Distributions of the peak frequency for $N_{\mathrm{iter}} = 3000$ sampled datasets plotted as a histogram. (a) Spin frequency f_{spin}; (b) Orbital frequency f_{orb}. The solid curve shows an ideal Gaussian distribution and the dotted vertical line in the middle marks the mean peak frequency $\bar{f}$. The shaded area in each histogram shows the $\pm 1\sigma$ interval which is the quoted uncertainty.

To express the discrepancy between the measured orbital and spin frequency from their corresponding literature values f_{lit}, the equation below was used to express the discrepancy in units of the 1σ uncertainty.

$$Discrepancy = \frac{|f_0 - f_{\mathrm{lit}}|}{\sigma_{f_0}}, \tag{12}$$

as quoted in Table II.

SCIENTIFIC SUMMARY FOR A GENERAL AUDIENCE

Most stars you will see in the night sky are born together with other stars and they evolved together for billions of years with gravity. Sometimes, one star burns its fuel faster than its companion then collapses into an extremely dense object called a white dwarf. If the two stars are close enough, the white-dwarf starts stealing material from the other star and the stolen material spirals around the white dwarf trying to fall into it, forming a glowing disc that shines brightly across the universe.

Some white-dwarfs have magnetic fields that are so powerful that the stolen material follows the field and hits the magnetic poles, illuminating them. Astronomers can observe distinct periodic lights from these processes as the stars move: a slower signal from the two stars orbiting each other and a faster one from white-dwarf spinning on its own.

In this investigation, we pointed our Durham telescopes at a magnetic 2-star system called MU-Camelopardalis to measure these signals. We measured both periods successfully to a fraction of their known values. Such astronomy techniques lets us understand the physics behind distant stars and they underpin the same ideas used in medical imaging, gravitational wave detection and exoplanet search.

GENERATIVE AI STATEMENT

Generative AI (Google's Gemini) was used to help write and debug Python code when producing high-quality, merged figures, including the two-panel light curve, the three-panel LS periodogram with false-alarm probability thresholds, and the bootstrap histogram figure. AI was also used to help to correct complications in the Monte Carlo bootstrap for loop and advise on the use of matplotlib parameters to meet the guidlines like correct axis labels, no gridlines, accessible line styles etc. Furthermore, AI was used to assist me in writing the Latex code with giving me the correct Latex syntax for table formatting, equations and custom commands.

Lastly, AI was used to evaluate my final draft of the lab reports where I needed to cut down on words so I used it to identify passages where information was repeated, where certain explanations were unnecessary for the given section. This allowed me to rephrase my explanations and meet the page limit. AI was also used to check the final report draft against the Durham Physics departmental guidelines where it led me to correct mistakes on figure formatting, style of referencing and error appendix upon identification of the errors.